Design patterns in a Quarkus bank API (A15)

- Part 1: 7 patterns found (DAO, DTO, Service Layer, Transaction Script, Controller, DI, Exception Translation); 21 concrete classes
- Part 2: Strategy, Observer, Domain Model added; types $7 \rightarrow 8 \rightarrow 9 \rightarrow 9$
- Part 3: 10 instances $\times$ Java/Python/JS/C++, 40/40 snippets pass all cases
- Median process time: C++ 3–4 ms, Python 15–22, JS 33–50, Java 79–124 (shared container)
- Not verified: real Quarkus build and DB transactions